~~x_1 x_2 y~~

	x_0	x_1	x_2	y
$x^{(0)}$	1	1	2	3
$x^{(1)}$	1	2	1	4
$x^{(2)}$	1	3	3	5

$$\theta \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}_{1\times d}$$

1] Problem 1 adding intercept

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$\boxed{0\,[0][0]+0[0][1]+0[0][2]}$

row number $= n$

$h(x) = 0[0][0] + 0 * x[0][1] + 0 * x[0][2]$

$= 0 * x[1][0] + 0 * x[1][1] + 0 * x[1][2]$

$= 0 * x[2][0] + 0 * x[2][1] + 0 * x[2][2]$

$$\theta_0 = \theta_0 - \alpha \boxed{\sum_{i=1}^{n} (h_0(x^{(i)}) - y^{(i)})\, x_0^{(i)}}$$

$$\boxed{\theta_0 = \theta_0 - \alpha \sum_{i=1}^{n} (h_0(x^{(i)} - y^{(i)})\, x_0^{(i)}}$$

$J(\theta) = \frac{1}{2} \sum^{n}$

Derivative function

$h_\theta(x^{(i)}) - y^{(i)}$

x_0	x_1	x_3
1	1	2
1	2	1
1	3	3

Parameter of function

$h_\theta(x^{(i)})$

$y^{(i)}$

$\theta_0 \rightarrow$ $x_0^{(1)} \leftarrow x[0][0]$

$x_0^{(2)} \leftarrow x[1][0]$

$x_0^{(3)} \leftarrow x[2][0]$

$\theta_1 \rightarrow$ $x_1^{(0)} \leftarrow x[0][1]$

$x_1^{(1)} \leftarrow x[1][1]$

$x_1^{(2)} \leftarrow x[2][1]$

θ_2 $x_2^{(0)} \leftarrow x[0][2]$

$x_2^{(1)} \leftarrow x[1][2]$

$x_2^{(2)} \leftarrow x[2][2]$

$$\theta_0 = \theta_0 - \alpha \sum_{i=1}^{n} (h_\theta(x_1^{(i)}) - y^{(i)})\, x_0^{(i)}$$

x_0

Cost function

hypothesis

Derivative

(θ_0)

$\theta_0 - \alpha \sum \boxed{0.001}$

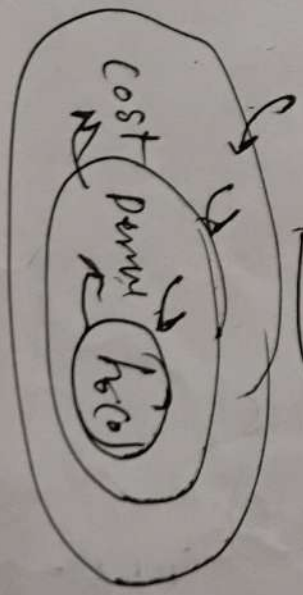
$$\theta_0 = \theta_0 - \alpha \sum_{i=1}^{n} (h_\theta(x_1^{(i)}) - y^{(i)})\, x_0^{(i)}$$

cost function

$$\sum_{i=1}^{n} (h_\theta(x_1^{(i)}) - y^{(i)})\, x_0^{(i)}$$

derivative

$(\theta_0 x_0)$



$$h_\theta(x) = \theta_0 x_0 + \theta_1 + x_1 + \theta_2 + x_0$$

$J_\theta > J_\theta$ (current)

0.01

model fid

Group

for i in range (len(x)) ← rows

$$\sum h_\theta(x^{(i)} - y^{(i)})\, x_0^{(i)} \qquad \underline{\underline{X_2^{(0)}}}$$

$(h_\theta^{(i)})$

Modification for coding

$$h_\theta^{(i)} - (x_{(0)}^{(i)} - y^{(i)})\, x_0^{(i)}$$

rows, cols

$$\cancel{(x_{(0)}^{(i)} - y^{(i)})}$$

$$\boxed{h_\theta(x^{(i)}) - y^{(i)})\, x_0^{(i)}}$$

$\theta_0 =$

$$\left[\left[\theta_0 x_0^{(0)} + \theta_1 x_{(1)}^{(0)} + \theta_2 x_{(2)}^{(0)}\right] - y^{(0)}\right] \times x_0^{(0)}$$

↱ x, theta, s = 1

$$\left[\theta_0 x_0^{(1)} + \theta_1 x_{(1)}^{(1)} + \theta_2 x_{(2)}^{(1)}\right] - y^{(1)}\right] \times x_0^{(1)}$$

x, theta, s = 2

$$\left[\theta_0 x_0^{(2)} + \theta_1 x_1^{(2)} + \theta_2 x_2^{(2)}\right] - y^{(2)}\right] \times x_0^{(2)}$$

Expand

↱ This parameter take from out side [ ] for θ_0

Samples what is sample [ ]

len(x)

[0] [0]

[1] [0]

[2] [0]

[1 × 1 + 2)

$[0] - 3)\,1 + (0 - 4)\,2 + (0 - 5)\,3$

$= 0 - 0.1(-3 - 8 - 15) \quad (-26)$

$= 0 - 0.1$

$1.2 - 3$

$-7.8 + 8 - 15 \quad -24.8$

$\boxed{-2.4.8}$

$\theta_1 = 0.41 =$

$\theta_2 = 0.85$

$\theta_3 = 0.76$

1 1.2 = 3

(2) = 4 tm

(3) 5 (lists)

$1 \times 0.41 + 4 \times 0.85 + 3 \times 0.76$

$5.27 \simeq 94\% \simeq 95\%$